

PRASANNA NAGARALE

Email: nagaraleprasanna@gmail.com | Phone: +91 9448178417 | Location: Bangalore, Karnataka
LinkedIn: [linkedin.com/in/prasanna-ai](https://www.linkedin.com/in/prasanna-ai) | GitHub: github.com/prasanna-nagarale | Portfolio: prasanna-nagarale.github.io/prasanna-portfolio

PROFESSIONAL SUMMARY

GenAI Engineer specializing in RAG systems, LLM fine-tuning, and multi-agent AI with production experience serving 500+ users. Expertise in building end-to-end intelligent systems using LangChain, FastAPI, transformer models, and computer vision. Proven results: 99.3% ML accuracy, 92% GenAI model performance, sub-2s response times across 8 deployed AI projects.

TECHNICAL SKILLS

GenAI & LLMs:	RAG, Fine-tuning, Prompt Engineering, LangChain, LangGraph, Hugging Face, Groq, Multi-Agent Systems
Machine Learning:	PyTorch, TensorFlow, Deep Learning, NLP, Computer Vision, Scikit-learn, Keras
Development:	Python, FastAPI, Django, Flask, Streamlit, PostgreSQL, Docker, Git, REST APIs
AI Frameworks:	OpenCV, FAISS, Transformers, Sentence-Transformers, Beautiful Soup, Pandas, NumPy
Data Science:	Feature Engineering, Model Evaluation, Data Visualization, EDA, Statistical Analysis

PROFESSIONAL EXPERIENCE

Software Development Intern - AI/ML Specialization

Feb 2025 - Jul 2025

ClickCrawl Media Pvt. Ltd., Hubli, Karnataka

- Engineered WordRift AI, a production GenAI platform serving 500+ users with RAG-powered features (paraphrasing, grammar correction, translation, AI detection) using transformer models (T5, BART, RoBERTa); achieved 92% accuracy and reduced response time 75% through LLM optimization
- Developed HDR image enhancement system using OpenCV and computer vision techniques for automated multi-exposure processing; delivered 90% professional-grade quality with 60% noise reduction in 2-second inference time
- Built scalable user management system with FastAPI backend, PostgreSQL database, premium tier logic, and real-time analytics dashboard for production ML monitoring

KEY PROJECTS

WordRift AI - Production GenAI Writing Platform [\[Live: wordriftai.com\]](https://wordriftai.com)

Full-stack AI writing platform with 11 integrated features serving 500+ active users. Implements transformer-based NLP pipeline (T5, BART, RoBERTa) for paraphrasing, grammar correction, translation (50+ languages), plagiarism detection, and AI humanization. Built with Django, PostgreSQL, Celery. Results: 92% semantic accuracy, 93% grammar correction rate, sub-2.5s processing time.

QueryFox AI - Advanced RAG System with Agentic Routing [\[GitHub\]](#)

Production-ready document intelligence platform combining RAG with LangGraph-powered agentic web search routing. Built with FastAPI backend, Streamlit frontend, and Docker containerization. Features multi-format document processing (PDF, DOCX, CSV), FAISS vector search, Groq LLM inference (Llama 3.3 70B), Tavily API integration. Tech: LangChain, Sentence-Transformers, Beautiful Soup. Metrics: <2s response time, 5+ file formats.

HDR Image Enhancement System - Computer Vision Pipeline [\[GitHub\]](#)

Automated OpenCV-based system for multi-exposure image merging using Debevec algorithm with tone mapping. Features batch processing, Django web UI with drag-and-drop upload, real-time preview. Performance: 90% professional-grade quality match, 60% noise reduction, 2-second processing time for 3 RAW images.

Crop Recommendation System - ML Classification [\[GitHub\]](#)

ML system achieving 99.3% accuracy using Random Forest and ensemble methods. Analyzes soil nutrients (NPK), pH, rainfall, temperature to recommend optimal crops from 22 varieties. Trained on 2,200+ samples with 0.98 AUC-ROC score. Deployed via Flask web interface.

Additional Achievement: Fine-tuned Phi-2 (2.7B params) for financial sentiment analysis using QLoRA; published on Hugging Face

EDUCATION

Bachelor of Engineering in Artificial Intelligence & Machine Learning

Dec 2021 - June 2025

Basaveshwar Engineering College, Bagalkot, Karnataka

CERTIFICATIONS

Career Essentials in Generative AI - Microsoft & LinkedIn Learning
Machine Learning with Python - IBM Cognitive Class
Introduction to Natural Language Processing - Infosys Springboard
Introduction to Artificial Intelligence - Infosys Springboard

LANGUAGES

English, Kannada, Hindi